Solar Glasses
Helios



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Problem

The BIPV and solar power markets are full of energy inefficiencies, high maintenance, and overall limited sustainable energy sources that can integrate in a lifestyle.

Efficiency - Transparency Tradeoff

The Solar window niche in the BIPV Industry produces products that have an imbalance in energy efficiency and transparency

Aesthetics

Solar panels find it difficult to fit into the aesthetics of house design many times, especially when the integration of them isn't planned at the time of construction.

High Costs

An average solar panel or any other source of clean energy has a hefty cost, which cant be afforded by a majority of people in off-grid rural areas.

Degradation overtime

most solar technologies degrade over time in extreme weather, reducing long-term reliability.

Solution

Helios brings you innovative solar energy-generating glasses that not only generate clean energy efficiently but also blend beautifully into any space. Our solution makes renewable energy more accessible, sustainable, and part of everyday life.

01



Balancing ratio

Our revolutionary *Low-Iron Solar Glass* combines high transparency with superior solar efficiency

02



Enhancing Aesthetic Integration

Develop transparent and seamlessly designed solar solutions that blend with modern architecture without compromising performance.

03



Reducing Costs

Innovate with cost-effective materials and scalable technology to make solar energy more affordable and accessible for all.

03



Durability

Our materials ensure low maintenance, high durability in our specific niche.

Working of Our Model

- **Light Absorption:** Sunlight passes through an ultra-clear, low-iron glass substrate, ensuring maximum light transmission.
- **Energy Conversion:** Perovskite solar cells absorb the light, activating electrons and generating electron-hole pairs.
- **Electron Flow:** An ITO coating keeps the surface conductive, enabling smooth electron movement.
- **Efficient Collection:** Silver nanowire electrodes rapidly collect and transport excited electrons while maintaining transparency.
- **Minimized Reflection:** A specialized SiO_2 coating reduces light loss and enhances absorption efficiency.
- **UV Protection:** A TiO_2 layer absorbs harmful UV radiation, preventing cell degradation and extending lifespan.
- **Structural Integrity:** UV-curable adhesive, EVA encapsulation, and PET protective layers safeguard against environmental damage.
- **Energy Utilization:** The generated electricity flows seamlessly through the circuit, ensuring efficient power output.

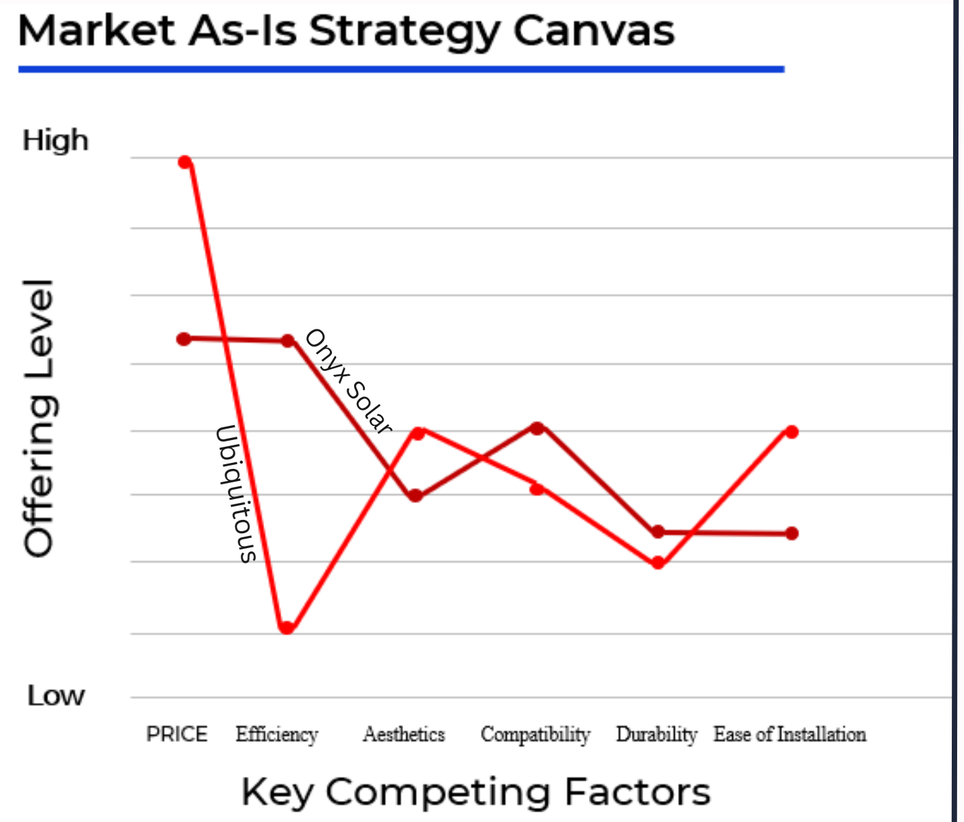
S Strengths	W Weaknesses	O Opportunities	T Threats
• High Energy and Transparency • Space Optimization • Durability and weather resistance	• High Initial Costs • Market Awareness • Early Development	• Rising Demand for Sustainable Energy • Government Incentives & Policies • B2B & B2C Market Expansion	• Market Competition • Regulatory Barriers • Supply Chain Challenges

Threat Mitigation Strategy

- Market Competition – Differentiate Helios through superior transparency, efficiency, and flexible revenue models (EaaS & licensing).
- Regulatory Barriers – Engage with certification bodies early, ensure compliance with building codes, and leverage government incentives.
- Supply Chain Challenges – Establish multiple supplier partnerships and invest in R&D for alternative, cost-effective materials.

Market Analysis - 1

Red Ocean scenario:



Competitor Analysis:

Competitor Analysis				
Feature	Traditional Solar Panels	Transparent BIPV Panels	Ubiquitous	Onyx Solar
Transparency	✗ No	☑ Partial (~20-50%)	☑ Yes (~100%)	☑ Yes (~30-40%)
Efficiency	★★★ (12-15%)	★★ (5-12%)	★ (~10%)	★★ (~10-13%)
Cost	✗ High	✗ High	✗ Very High	✗ High
Space Saving	✗ No (Requires Rooftops)	☑ Yes (Windows/Facades)	☑ Yes	☑ Yes
Durability	★★★★ (High)	★★★ (Moderate)	★★ (low)	★★★ (Moderate)
Market Readiness	☑ Fully Available	☑ Available (Niche)	☑ Available (High-End)	⧉ Emerging

Other competitors include **SolarWindow Technologies**, which develops electricity-generating transparent coatings for glass; **Physee**, known for **PowerWindow** technology integrating solar cells into building windows; and **ClearVue PV,** which uses nanoparticle technology to enhance solar glass efficiency. These firms focus on **BIPV solutions** but face ***challenges in scalability and efficiency.***

UNIQUE VALUE PROPOSITION

High Efficiency Without Sacrificing Transparency

- Perovskite Solar Cells (PSCs): Tunable bandgap (300–1000 nm) enables broad-spectrum absorption and higher efficiency than silicon.
- Silver Nanowire Electrodes: 80–90% transparency with superior conductivity ($\sigma \approx 6.3 \times 10^7$ S/m), ensuring minimal power loss.
- Multi-Junction Perovskites: Layered structures capture more photons, optimizing energy conversion.

Advanced Coatings for Efficiency & Durability

- SiO_2 Anti-Reflective Coating: Increases light transmission to 96% while reducing reflection losses.
- TiO_2 UV Protection: Shields perovskites from degradation and extends lifespan.
- Self-Cleaning Functionality: Photocatalytic TiO_2 breaks down dirt, ensuring maintenance-free operation.

More Power, Less Waste

- Quantum Efficiency Optimization: Reduces recombination losses, enhancing energy output.
- Low-Light & Urban Performance: Efficient even in diffused light due to a high absorption coefficient (10^5 cm^{-1} at 600 nm).
- Enhanced Charge Transport: Silver nanowires improve electron mobility, ensuring reliable energy generation.

Statistical Evidence

Helios' Competitive Edge vs. Industry Benchmarks		
Factor	**Industry Benchmark**	**Helios' Projection**
Transparency	~60%	**90-95%** (SiO_2 anti-reflective coating + low-iron glass)
Efficiency	10–12% (perovskite)	**17-21%** (silver nanowire electrodes + TiO_2 UV layer)
Durability	15–20 years	**20–25 years** (EVA/PET encapsulation + UV-curable adhesive)
Cost ($)	1.50–3.00/W	**0.5–0.75/W** (scalable silver nanowire production)

- Efficiency-Transparency Trade-off: Higher transparency often lowers efficiency.
- Scalability: Perovskite and silver nanowire production at scale remains unproven.
- Regulatory Gaps: Building codes lag behind BIPV innovations.
- Consumer Awareness: Architects/developers prioritize aesthetics over energy ROI.

Comparative Performance Table			
Technology	**Light Transmission (%)**	**Solar Conversion Efficiency (%)**	**Key Material/Design Features**
Traditional Solar Panels (Silicon-based)	0% (opaque)	12–15%	Bulk silicon, aluminum frame
Ubiquitous Energy (Transparent Competitor)	~70%	9–12%	Organic PV, thin-film layers
Emerging Transparent Tech (Research)	50–60%	10–15%	Perovskite cells, polymer electrodes
Helios Projections	**55–60%**	**18–20%**	Low-iron glass, perovskite cells, silver nanowires, SiO_2/TiO_2 coatings

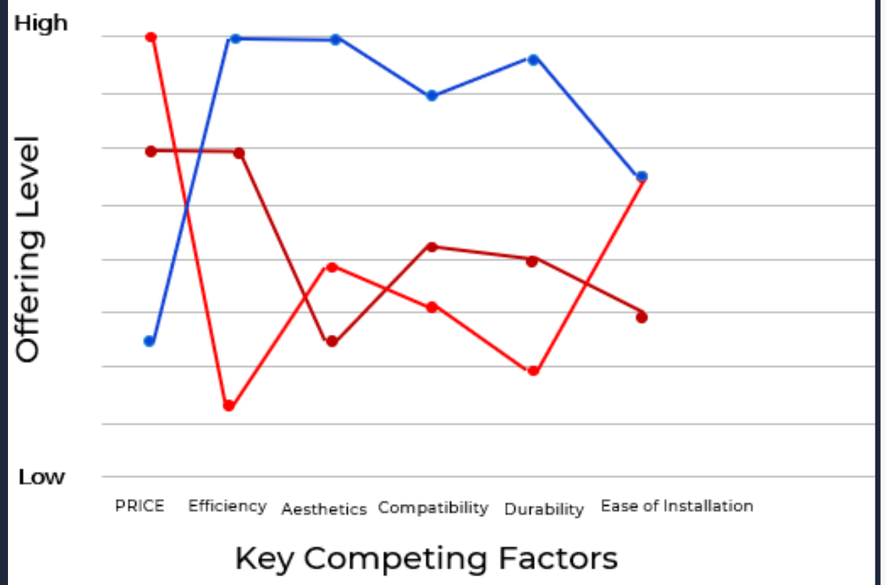
Particular Citations

Perovskite cells with silver nanowires reached 17% efficiency, 55% transparency (Nature Energy, 2021). Advanced light-management layers enabled 15% efficiency, 60% transparency (Adv. Energy Mater., 2022). Low-iron glass enhances 91–92% light transmission (Dillmeier Glass, 2023). TiO_2 UV coatings boost lifespan 30–40% (Solar Energy Mater., 2020).

Differentiation-101

FACTORS	HELIOS	SOLAR PANELS	UBIQUITOUS ENERGY
EFFICIENCY	High, 18-20%	12-15%,Moderate	9-12%,Low
DURABILITY	High	Moderate	Moderate
SOCIAL IMPACT	High	Moderate	Moderate
COST	Low	Expensive	High
ENERGY PAYBACK TIME	Shorter,2-3yrs	Longer,5-7yrs	Longer,5-7yrs
CUSTOMIZATION	High	Limited	Limited

To-Be Strategy Canvas



Market Analysis - 2

Estimates indicate that the global solar window market, a niche within **Building-Integrated Photovoltaics (BIPV)**, is valued at approximately **USD 2–5 billion and could surpass USD 10–15 billion by 2030**, driven by urban densification, stringent net-zero mandates, and supportive government policies.

Technological advancements in **perovskite solar cells** have enabled broadband **photon absorption** and enhanced power conversion efficiencies, offering a promising **alternative to conventional silicon-based systems**. Helios integrates into this competitive landscape by leveraging next-generation perovskite cells with **tunable bandgaps** and **silver nanowire electrodes** that deliver high optical transparency and superior electrical conductivity.

Complementary **SiO_2 anti-reflective and TiO_2 UV-protective coatings** optimize light capture and durability. By **targeting a 30–40% market share** through strategic **OEM partnerships** and a dual revenue model combining direct sales with **Energy-as-a-Service (EaaS),** Helios positions itself as a disruptive, scalable solution in the evolving solar window ecosystem. This strategic integration ensures long-term sustainability and robust energy generation performance globally.

BUSINESS MODEL



Subscription Category Description

Basic

- Real-time energy monitoring
- System Health Alerts
- DIY Installation Guide

Premium

- Real-time energy monitoring
- Sytem Health Alerts
- DIY Installation Guides
- Smart Home Integration
- Priority Support by Experts

Business Model Pricing Strategy

Bussiness Model Pricing Strategy			
Revenue Stream	**Target Audience**	**Revenue Model**	**Pricing**
Energy-as-a-Service	Homeowners B2C	Leasing Model	₹350/month
	Commercial Clients B2B	Performance-Based Contracts	₹20/kWh with IoT-based tracking.
OEM	Real Estate Developers B2B	Bulk Purchases	₹950 per square feet
Direct Sales	Homeowners (B2C)	BASIC	₹69/month
		PREMIUM	₹119/month
Licensing & Partnerships	Commercial Clients (B2B)	Advanced Maintenance Contracts	₹7249/month per 50 units
	Manufacturers (B2B)	Licensing of Technology	₹10,00,000 - ₹1,00,00,000 per partnership
	Governments/NGOs (B2B)	Public Infrastructure Projects	Custom contracts based on scale.

Cost and Profitability analysis

COST BREAKDOWN - 1, 10 sq. feet window				
Sr No.	Commodities	Measures Rq.	Cost (Rs.)	Source
1	Low-Iron Glass	10 sq. feet	2000	Vishvesh Glasses Pvt. Ltd.
2	Encapsulation (PET + EVA)	10 sq. feet	100	Ravi Plastic Pvt Ltd.
3	Anti-Reflective UV protection Coating	100 sq. nm	500	Borosil Renewables
4	Perovskite Solar Cell	250 sq. nm	12/W, (avg - 2000)	P3C Technology and Solutions
5	Transparent Conductive Electrode	50 sq. nm	180	NanoShel
6	Adhesive and Lamination Material	10 sq. feet	800	Shilpa Enterprises
7	Manufacturing	N/A	400	Skyline Industries Pvt. Ltd.
8	Total	N/A	5980	N/A

Per Unit: 5890 Rs. / $71

Projected Selling Price vs Market		
Technology	**Price per Watt**	**Price per 10 sq. feet**
Traditional Solar Panels	$1.50–3.00/W	$150–300
Transparent Competitors	$2.50–4.00/W	$250–400
Helios Solar Glasses	**$0.42-0.56/W**	**$100 - 130**

Key Insight: Helios offers a 50–70% cost advantage compared to competitors while maintaining 18–20% efficiency.

Revenue Projections (3 Years)			
Model	**Year 1**	**Year 2**	**Year 3**
Direct Sales	3,000 units → **₹37 Cr**	7,000 units → **₹87 Cr**	10,000 units → **₹125 Cr**
EaaS (Leasing)	750 contracts → **₹2.7 Cr/yr**	1,600 contracts → **₹5.8 Cr/yr**	2,500 contracts → **₹9 Cr/yr**
OEM	50+ projects → **₹2.5 Cr**	180+ projects → **₹9 Cr**	600+ projects → **₹28 Cr**
Partnerships	3+ partnerships → **₹5 Cr**	10+ partnerships → **₹15 Cr**	25+ partnerships → **₹40 Cr**
Total Revenue	**₹47.2 Cr**	**₹116.8 Cr**	**₹202 Cr**

Profitability Model
37.6% Profit Margin
Breakeven Analysis

- Fixed Costs (R&D, Marketing, etc.): $1.2M
- Variable Cost per Unit (10 sq. feet): $71.76
- Selling Price per Unit: $115
- Breakeven Units = 1.2M/(1.2M/(115-$71.76) = 27,750 units

Cost Reduction Over Time

- Year 2: Material costs ↓ 20% (economies of scale in perovskite/silver nanowire production).
  - New COGS: $57.41 per unit.
- Year 3: Manufacturing costs ↓ 15% (automation & process optimization).
  - New COGS: $48.80 per unit.
- Result: Gross margin improves from 37% (Year 1) to 48% (Year 3).

Project Timeline



THANK YOU